

and include latest displays for day and night vision. Master MS2038C is a VNA and spectrum analyser in one unit. It can also be used as a power meter.

Site Master S820E is used as a cable and antenna analyser for verification of various components, like antenna, radar and cables, used in avionics communication systems.

Remote spectrum analyser MS27103A is used for interference testing, network monitoring and multi-carrier monitoring. It is IP67-compliant, and is suitable for outdoor installations and network monitoring.

LMR Master S412E is a land mobile radio modulation analyser. It supports P25, DMR, APCO, NNBFM, TETRA and other standards used in radio communication by the avionics industry. LMR Master S412E can be used for indoor signal coverage application, which is important in mission-critical applications. LMR Master is equally good for radio applications, since this comprises VNA, spectrum analyser and power meter in a single, portable instrument.

Tripathi adds, "USB-based power sensors are other popular instruments used for power measurements. We have updated our LMR Master S412E that has digital technologies including Long-Term Evolution (LTE). LTE is used for data-centric applications by industry. We have added external USB-based oscillators and other related products to make LMR Master a complete radio testing solution."

Joseph Pozzolano, vice president - sales and marketing, Behlman Electronics, says "We supply a number of 400Hz frequency converters to the aviation test industry. Our specialty is clean power in a rugged and reliable package. We manufacture a broad line of AC power supplies in both single-phase and three-phase output and frequency converters from 1000VA to 120,000VA. These are available with variable or fixed voltage and frequency output, manual or programmable control, with a variety of interfaces, unique protection



Fig. 2: VNA Master MS2038C
(Credit: www.anritsu.com)



Fig. 3: Remote spectrum monitor MS27102A
(Credit: www.anritsu.com)



Fig. 4: AC power supplies and frequency converters for aviation testing (www.behlman.com)

circuitry and the ability to modify units as per application."

Any worldwide utility can be simulated by Behlman's P series. One simply needs to select the required frequency, dial-up desired voltage and voltage range, and it is ready for testing. It provides clean sine-wave output with excellent line and load regulation, low harmonic distortion and high efficiency in a compact size.

BLHP series (Behlman's high power, AC power sources/frequency

converters/inverters) models come with variable voltage and frequency, external sync and 0V DC to 10V DC remote control. There is optional IEEE 488 or RS232 interface for computer control. High efficiency, small size and quiet operation make the BLHP series suitable for precision avionics testing requirements.

Their BL+ series provides linear performance with the benefits of a switching amplifier. It offers smaller footprint, reduced downtime and greater energy efficiency. It comes with an optional computer control for automatic test equipment applications. It provides adjustable frequency and voltage, allowing for maximum flexibility. FC5003 frequency converter is designed to convert 120V AC/208V AC, 3-phase, 60Hz ground power to 115V AC/200V AC, 3-phase, 400Hz aircraft power.

Other requirements

To establish certified aircraft effective perceived noise levels, aircraft noise certification tests are required at each approach in accordance with relevant aircraft noise certification standards. Standards include ICAO Annex 16 and/or FAR 36 and IEC 61265.

In the current market dynamics, there is a need to keep evolving technologies to enable the aviation industry to carry out more precise and realistic testing. T&M vendors need to stay ahead of the evolving technical requirements so that the right testing equipment is available when it is required.

Next-generation air transportation systems will require test capabilities for aircraft avionics systems. As the world moves from analogue, ground-based aircraft communications to satellite-based, digital aircraft communications, test requirements will change, requiring more digital instrumentation than analogue instrumentation. In essence, it will flip the switch on test requirements to a new set of completely different parameters. As new technologies are added to the field, test capabilities must be upgraded to stay ahead of those new technologies. **EFY**